

Teaching Plan

Term Apr 2026

Module Name BACK-END DEVELOPMENT

Credit Units 4

Module Leader TAN CHOON PENG

Contact 60 hrs

Wk	Begin Date	Topics	Activity/Session 1	Activity/Session 2	Assessment
1	20 Apr	Introduction to Backend Development - Explore common backend technologies - Front-end technologies vs Back-end technologies - Introduction to Node.js and Express.js - Install Node.js and npm - Setup a basic Express.js application	Module briefing Software installation	Getting started with Node.js and Express	
2	27 Apr	Building API using Node.js and Express - Introduction to API - Retrieve data using GET Request - Create data using POST Request - Update data using PUT Request - Delete data using DELETE Request - Accessing API using Postman	Practical on topic	OAL on topic	
3	04 May	Building API with Database Integration - Connecting to a relational database - Creating records in database - Retriving records from database - Updating records in database - Deleting records from database	Practical on topic	OAL on topic	
4	11 May	Building Maintainable & Secure APIs - Model-View-Controller (MVC) architecture - Server-side input data validation using middleware - Error handling using Try-Catch-Finally construct - Parameterized queries to prevent SQL Injections	Practical on topic	OAL on topic	
5	18 May	Whitespace Week - Integrating back-end APIs to front-back application	Practical on topic	OAL on topic	

		through JavaScript API calls.			
6	25 May	Managing Entities and Relationships in APIs - Construct complex SQLs with JOINS to retrieve JASON output data compiled from multiple tables	Practical on topic Assignment Briefing	OAL on topic	
7	01 Jun	Common Test Revision	Common Test Revision	Assignment Consultation	
8	08 Jun	Common Test	Common Test	Common Test	Common Test (30%)
9	15 Jun	Break			
10	22 Jun	Break			
11	29 Jun	User Authentication and Authorization - Introduction of JSON Web Token (JWT) - JWT generation during User Authentication - API Authorization with JWT	Practical on topic Assignment Consultation	OAL on topic	Assignment 1 (40%)
12	06 Jul	User Authentication and Authorization - API Authorization with JWT Token	Practical on topic	Assignment Consultation	Assignment 1 (40%)
13	13 Jul	Unit Testing & API Documentation - Implement unit tests with Jest framework for API controllers and models, including mocking dependencies - Generate API documentation with Swagger to understand input data requirement for testing of API end-points from web browser.	Practical on topic Assignment Consultation	OAL on topic Assignment Checkpoint submission	Assignment 1 (40%)
14	20 Jul	Whitespace Week Assignment Consultation	Assignment Consultation	Assignment Consultation	Assignment 1 (40%)
15	27 Jul	Summative Quiz and Assignment Consultation	Summative Quiz (15%)	Assignment Consultation Assignment Final Submission	Assignment 1 (40%)
16	03 Aug	Assignment Demo	Assignment Demo	Assignment Demo	Assignment 1 (40%)
17	10 Aug	Assignment Demo	Assignment Demo	Assignment Demo Assignment Reflection Report Submission	Assignment 1 (40%)
18	17 Aug	Study & Exam Week			
19	24 Aug	Study & Exam Week			

Holiday:

01 May - Labour Day
27 May - Hari Raya Haji
01 Jun - Vesak Day
10 Aug - National Day

Notes on Teaching Plan:**Textbooks / Recommended Readings:**

1. Explore Web Development with Node.js (<https://www.linkedin.com/learning/paths/explore-web-development-with-node-js>)
2. Express Essentials: Build Powerful Web Apps with Node.js (<https://www.linkedin.com/learning/express-essentials-build-powerful-web-apps-with-node-js>)
3. Building RESTful APIs with Node.js and Express (<https://www.linkedin.com/learning/building-restful-apis-with-node-js-and-express-16069959>)